



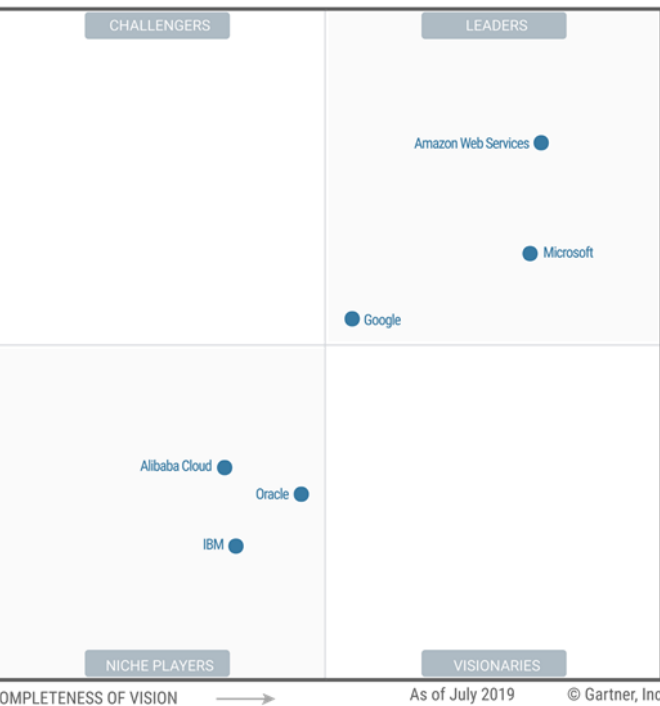
ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Modern IT challenges

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The top player [Gartner, 2019-2022]



Amazon AWS vs. MS Azure vs. Google cloud

- Amazon AWS is the leader. *Who would have guessed that an online book seller would lead the pack in offering IT infrastructures?*
- Microsoft's Azure is coming close to AWS in terms of offering especially thanks to the **Office 365 ecosystem**. At the beginning, some customers have complained of too many outages, leading Gartner to advise those running critical applications on Azure to have even some non-Azure backups. Problem solved today.
- While Google cloud has some top technologies, it is still in “the rudimentary stages of learning to engage with midmarket customers”. Its main business is more in the PaaS-SaaS context; it is less interested to the crowded IaaS arena



Why is MS Azure growing?

- Integration with desktop software
- Richness of applications ported or integrated with cloud
- *Reliability, scalability and quality of service* are assumed as guaranteed



2013: CIA's private cloud

- 18 months of work
- \$600 million deal for 4 years
- The new cloud will be rolled out across 17 different intelligence agencies



Amazon vs IBM

- The contract is for a lot of money, so it is quite attractive from a commercial point of view
- Even more important, it is the implicit endorsement → CIA choosing AWS provides Amazon a jolly card in all discussions about **security**, **trustworthiness**, **availability**, and not only



Why Amazon?

- 1) Amazon provides customers with ease of use and rapid availability of resources
- 2) Rich ecosystem of applications, tools and libraries provided by itself and many third parties (*even the CIA recognized that a richer environment provides several additional benefits*)
- 3) It has always been a company oriented to sell everything
- 4) It prefers a large set of cheaper components (servers, routers, storage arrays) instead of premium equipment, thus guaranteeing lower prices and focus on smart management and adaptable infrastructure



Motivations (1/4) – *customer orientation*

- Amazon has grown on **software developers** in addition to **system administrators** focused on IT operations. The motivations, judgment criteria, and agility expectations of the two groups are different, but AWS services were immediately orientated to:
 - **PaaS in addition to IaaS**
 - **ease of use and rapid availability of resources**
- Meanwhile, most of initial Amazon's competitors provide hosting services with limited self-service operations
- Most of its competitors used to sell IaaS operations instead of rich services



Motivations (2/4) – *rich ecosystem*

- The AWS ecosystem provides an enormous advantage for users, enabling them to deploy applications quickly → One of the main reasons developers embrace AWS is because of the **richness of its services**. This includes
 - services that AWS itself provides
 - a very large number of services provided by third parties
- Developers can integrate applications together by combining these services with their own business logic. From the perspective of a user, the richer the ecosystem, the better: a rich ecosystem provides time-to-market advantages, greater flexibility in terms of suppliers and application architecture choices, and lower costs through supplier competition
- The alternative for users with other cloud providers is to implement those services on their own



Motivations (3/4) – *many inexpensive components*

- Amazon pursues a different path in how it designs and operates its cloud environment
 - Most cloud providers tout the quality of the kit used to build their cloud: top name-brand servers, routers, storage arrays and so on
 - Amazon is notoriously cheap and refuses to pay premium prices for its equipment
- Amazon assumes that it will be operating its offering at vast scale and cannot afford to use designs that cannot grow to support that assumption. As an example, unlike most cloud providers, Amazon uses Layer 3 networking rather than Layer 2, because the latter ends up tied to VLAN topologies that do not scale
- It cannot have escaped the CIA's notice that the explosion of **big data** and next-generation applications is far better served by a smart, adaptable, agile infrastructure



Motivations (4/4)

The design approach of using several inexpensive hardware devices goes beyond just using inexpensive kit to save money

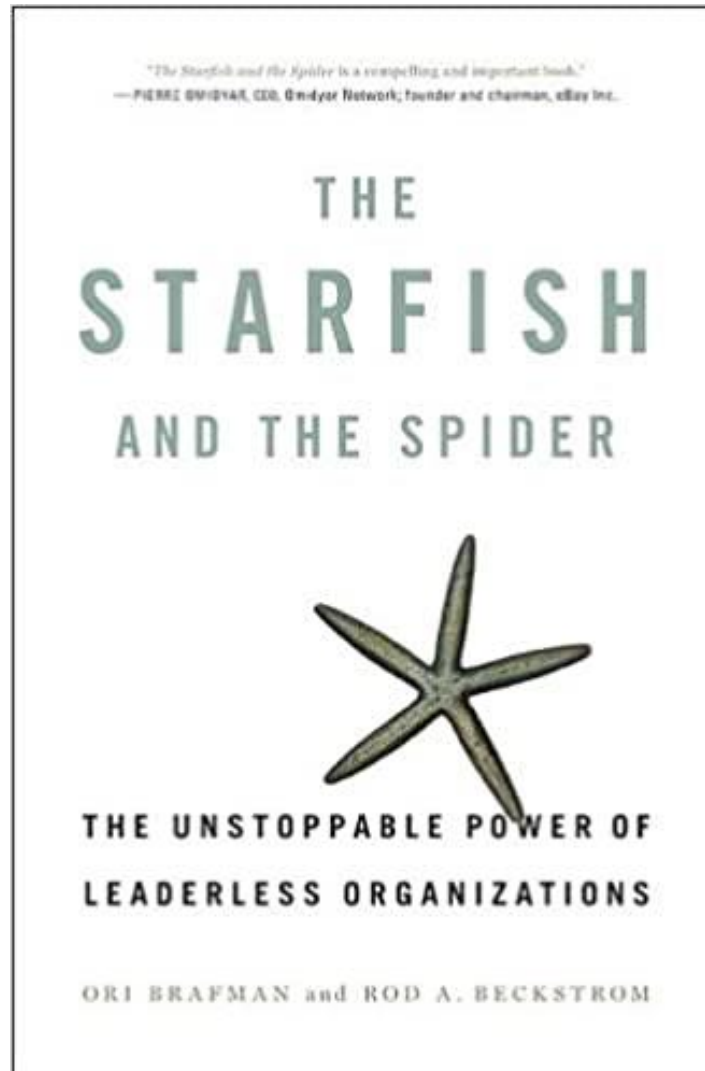
- It is driven by Amazon's recognition that, **at large scale, hardware fails constantly, no matter how cheap or expensive**. If you run a robust, highly available environment, then you cannot depend just on the quality of the underlying hardware, **but you must use redundancy to avoid hardware-caused service outage**
- makes it easier to create new services and applications
- and, even more important, it offers advantages in terms of **scalability**

This choice requires **sophisticated coordination** to ensure that:

- 1) there are sufficient redundant resources available
- 2) data is replicated to those resources
- 3) cloud services are operated on redundant devices, monitored and coordinate to avoid service unavailability



Reliable systems are distributed and autonomous



Modern workload



Definition: “Workload”

“The amount of work to be done”

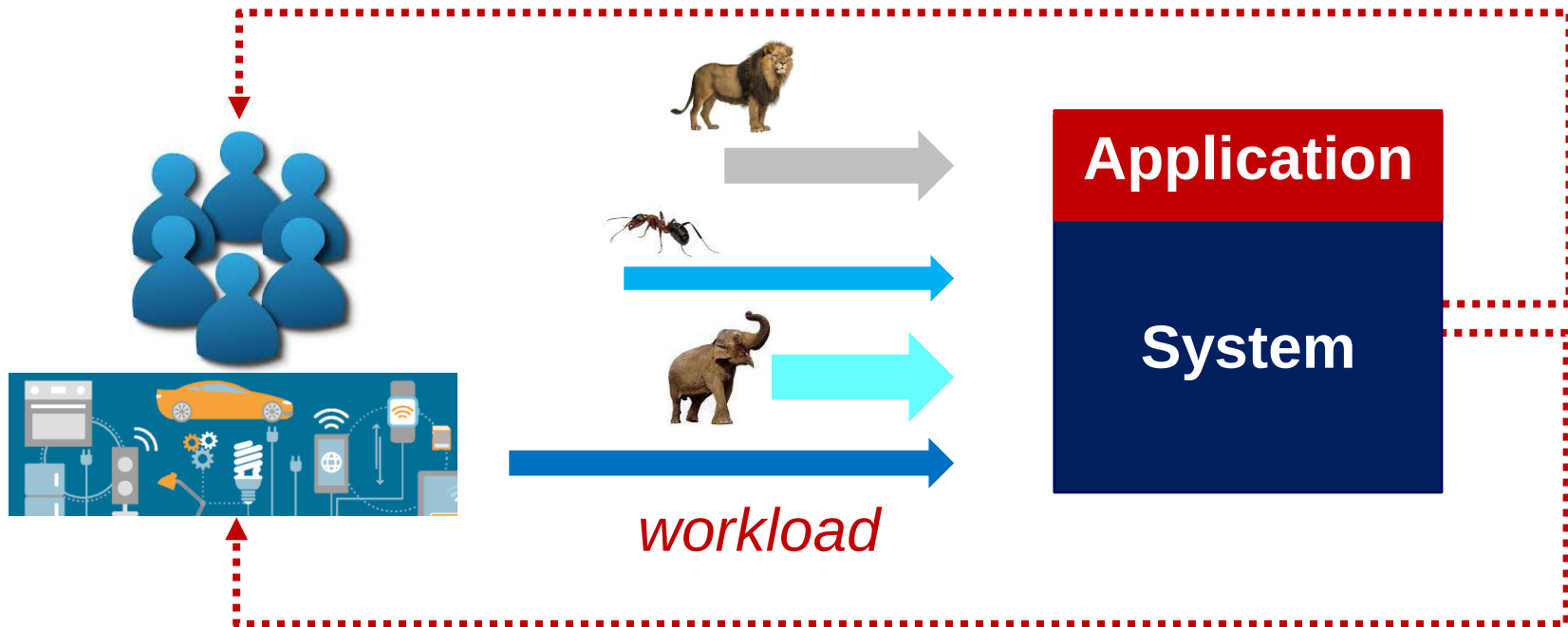
“The amount of processing that the computer system has been given to do at a given time”

Usually, the internal tasks are not considered in the workload, hence as **workload** we typically consider the **offered workload** →

“The amount and type of service requests expected from the external by some number of users and jobs connected to and interacting with the system in a specified time period”



Offered workload (*not only people*)



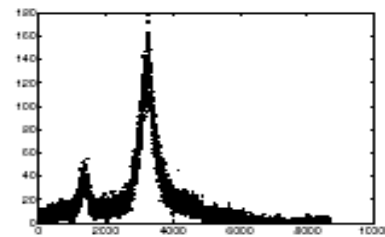
WWW opened the information systems to the world

Opening the services to the world population represents a huge issue for any design and system engineer

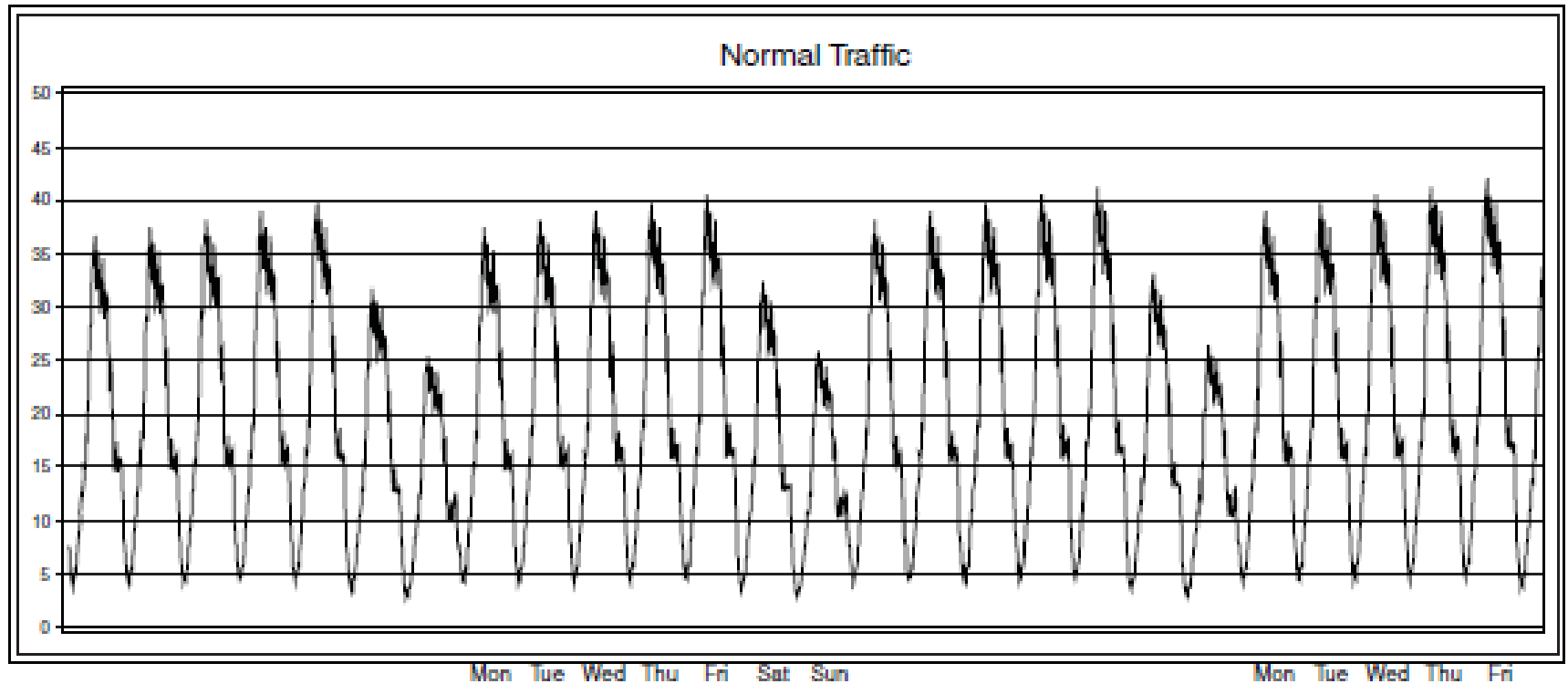


It is tough to be reliable when:

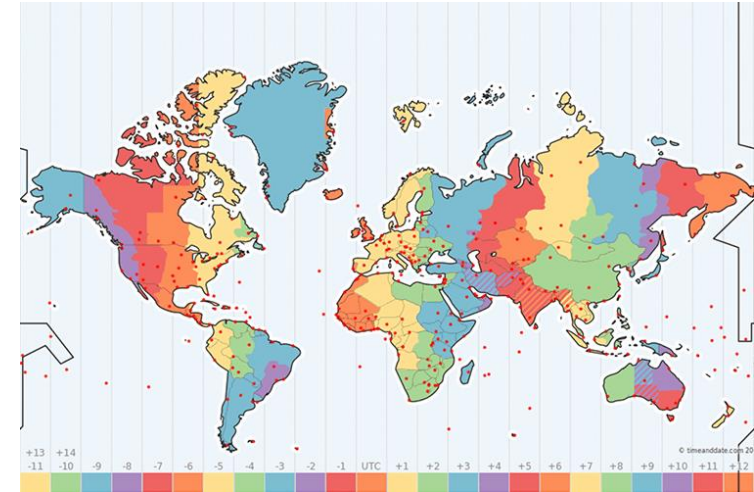
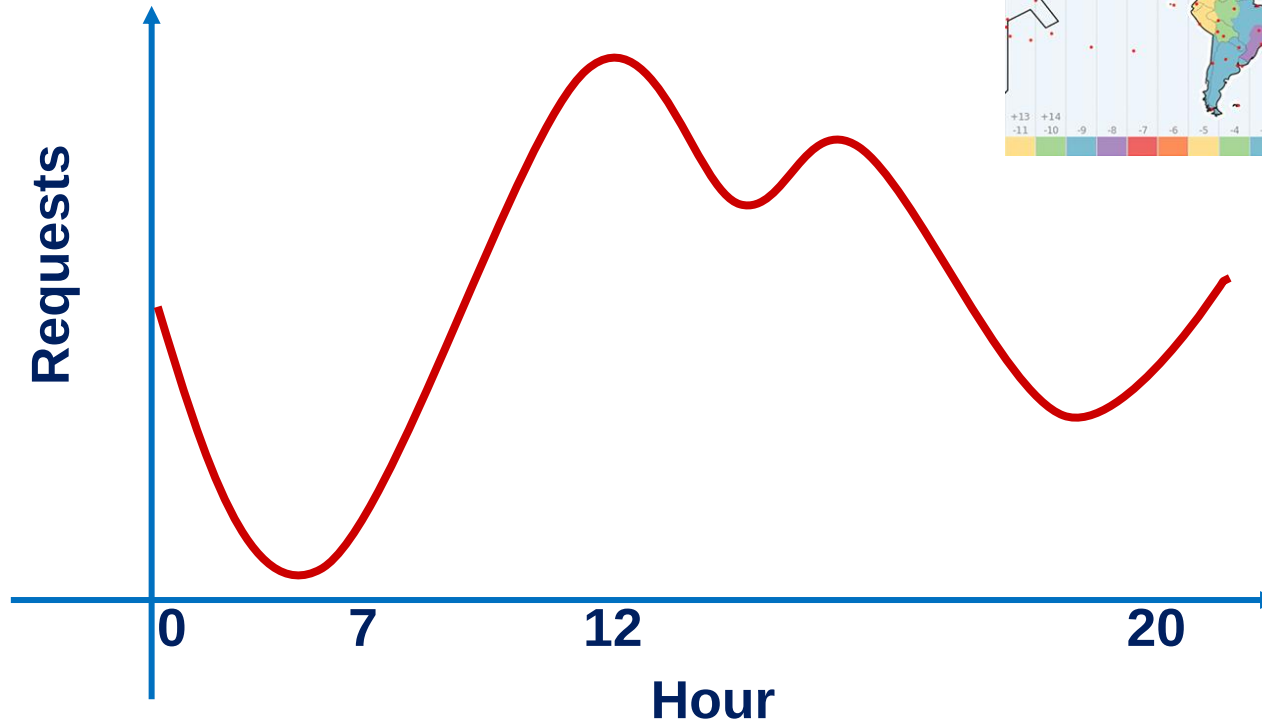
- you do not know how many users can reach your service
- you do not know the mix of requests
- service times behind each request may have different orders of magnitudes (from 1 msec to 1 sec)
- your system may be subject to hotspots, burst arrivals, etc.
- you may experiment enormous increase/decrease of requests per day, per month, per year



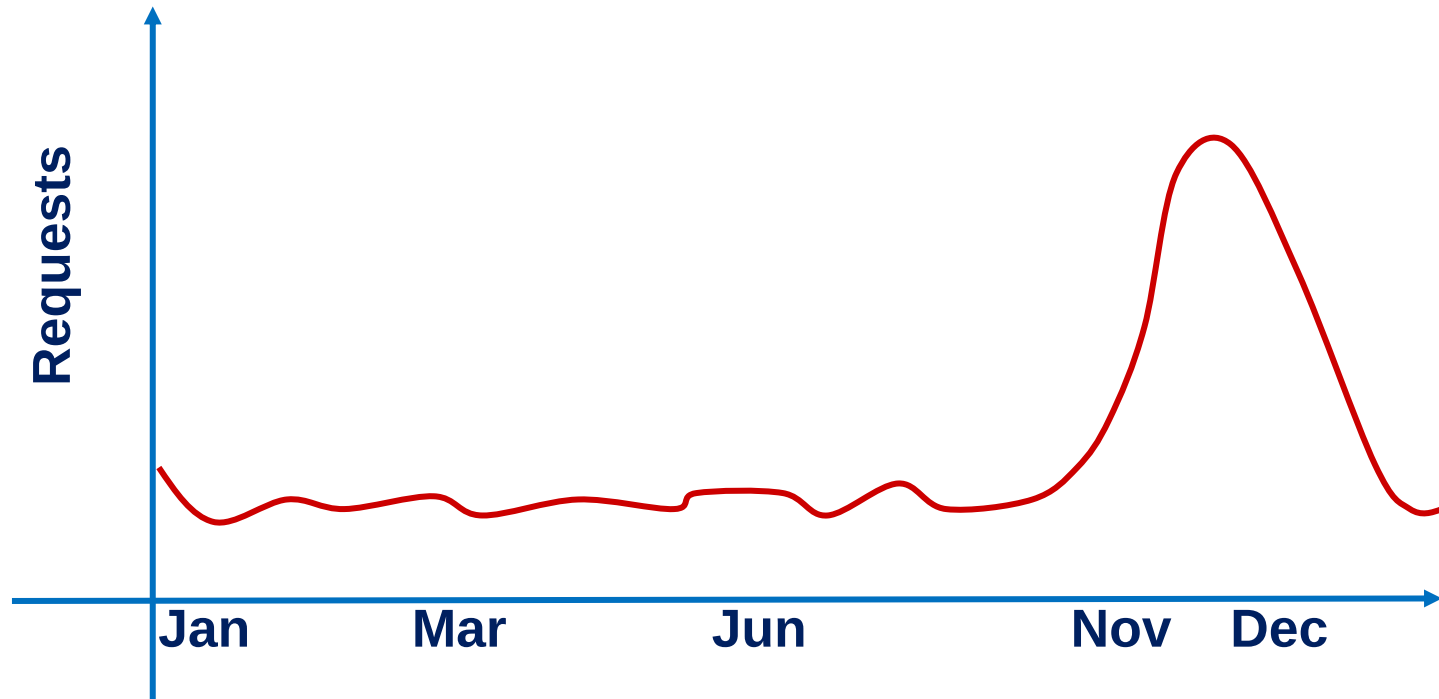
Typical weekly workload



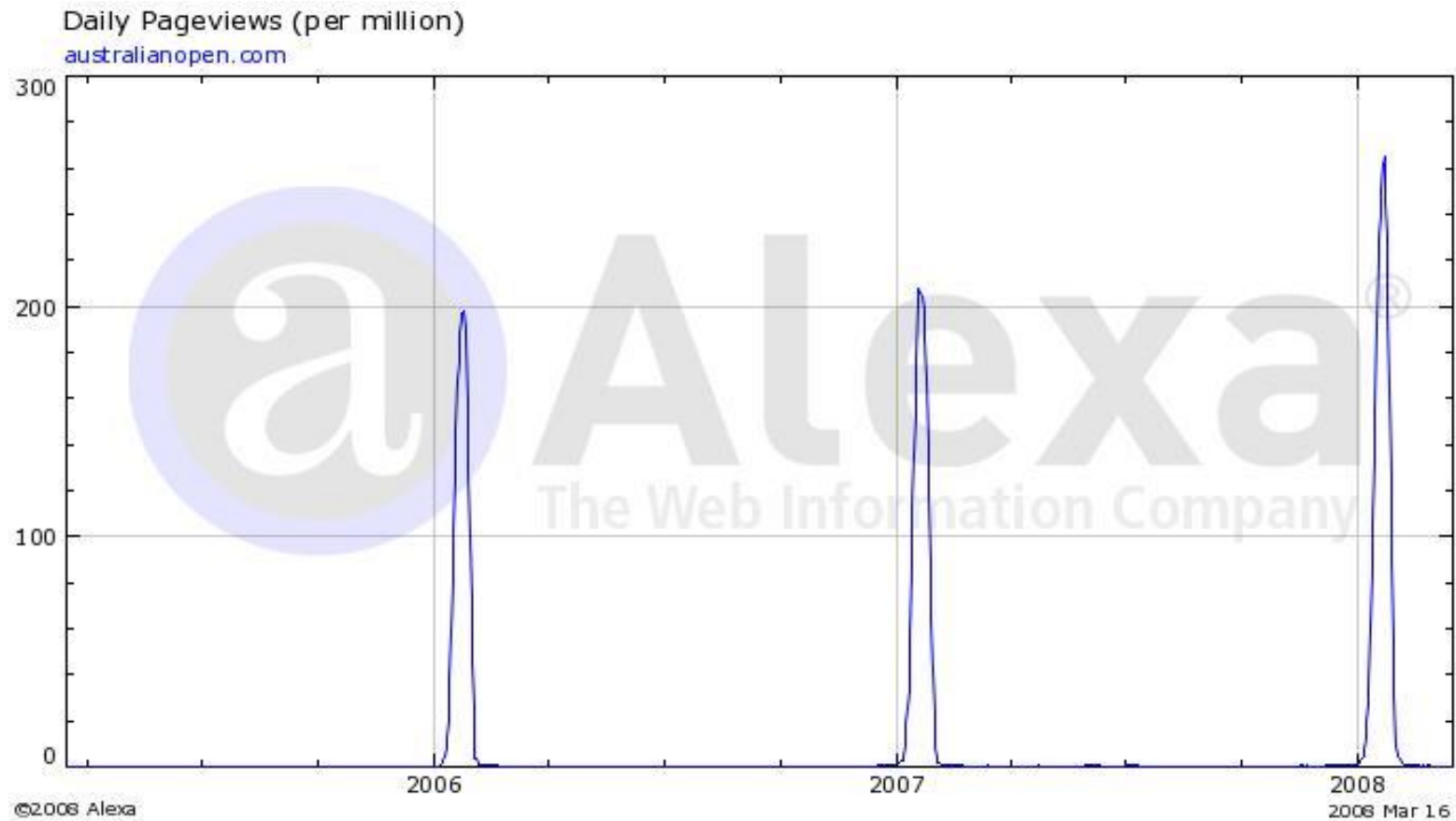
Typical daily workload (in each time zone)



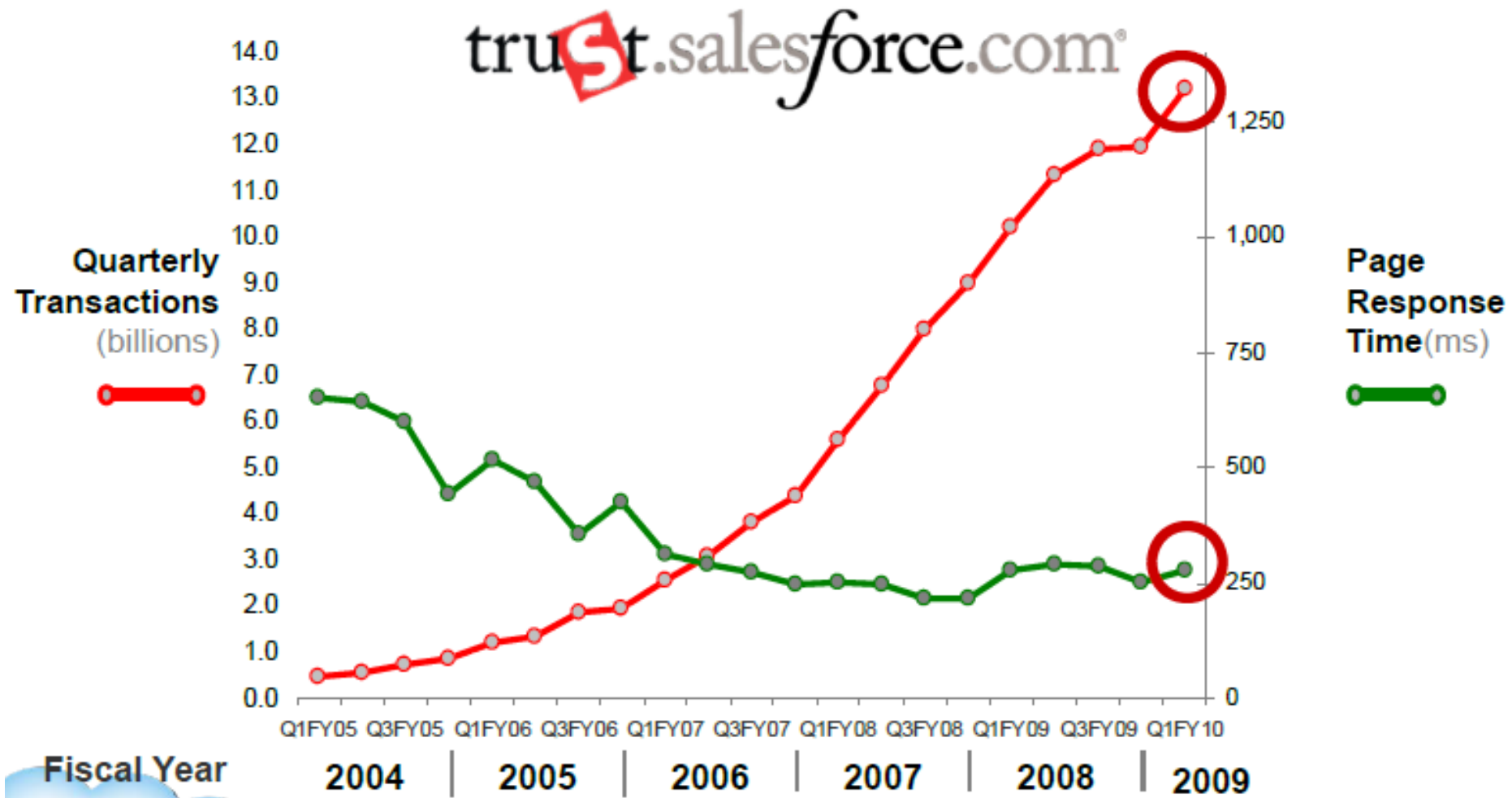
Seasonality of e-commerce



Periodic peaks of sport events (e.g., Australia Open)

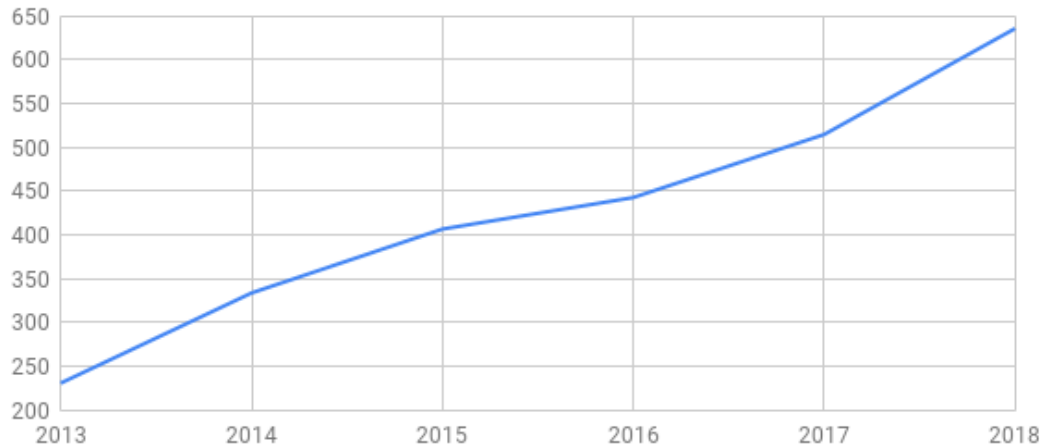


Saleforces (from 200K to 13 billion hits/quarter in 6 years)



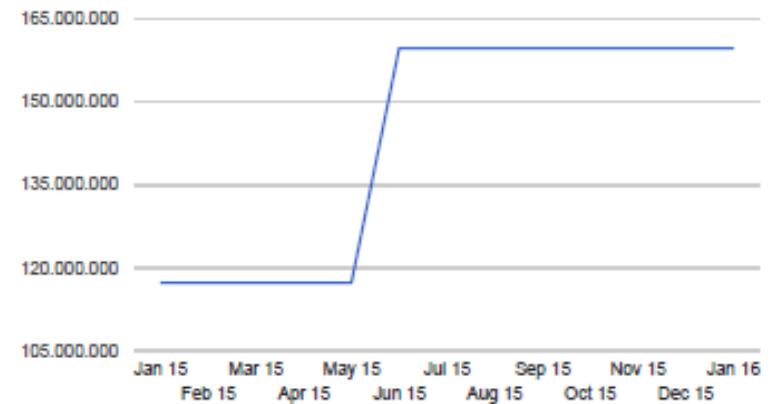
Number of visits to Alibaba.com

Alibaba Annual Active Customers | 2013-2018 | DMR (in millions)



Alibaba.com Traffic Estimate

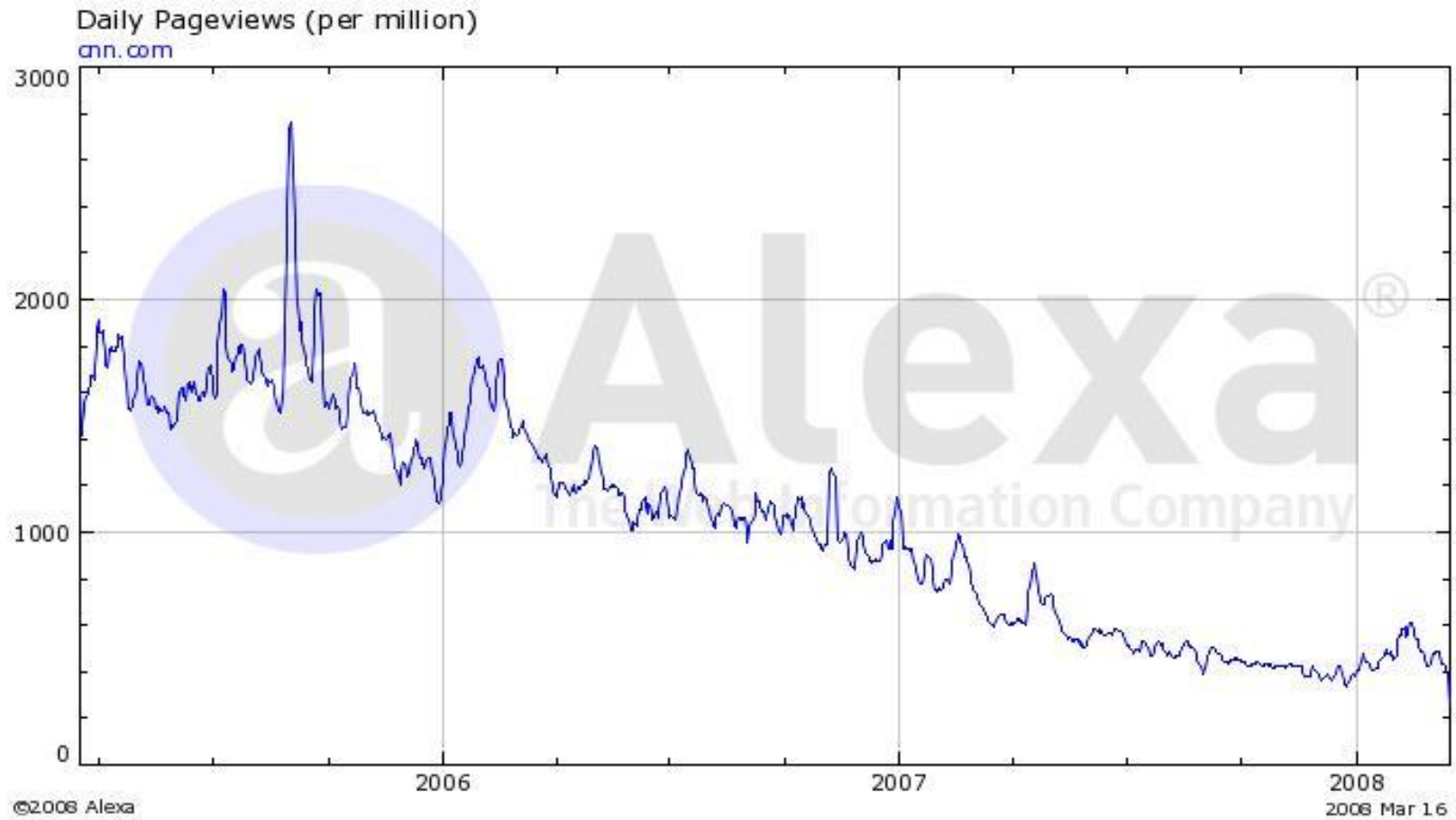
Estimated Monthly Traffic (visits) for Alibaba.com - By Month



Monthly visitor traffic is up 36.3% year over year



But you can experiment even enormous decreases (e.g., CNN from 2 to 1 billion hits/day)



Number of visits to eBay.com

Number of unique U.S. visitors to eBay.com from April 2011 to October 2014 (in millions)



Source:
Compete

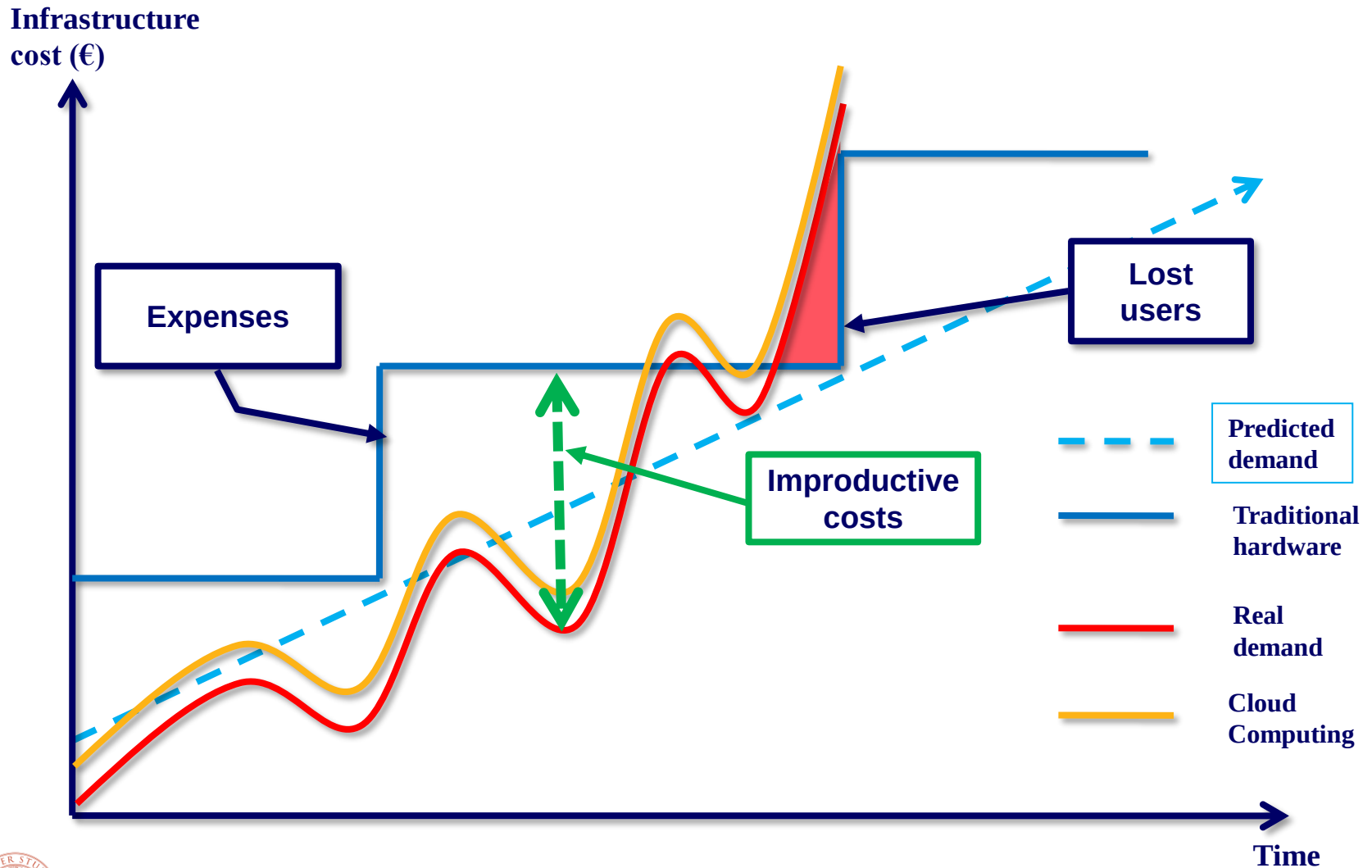
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Additional Information

United States; Compete; April 2011 to October 2014



Elastic sizing: An important motivation for cloud



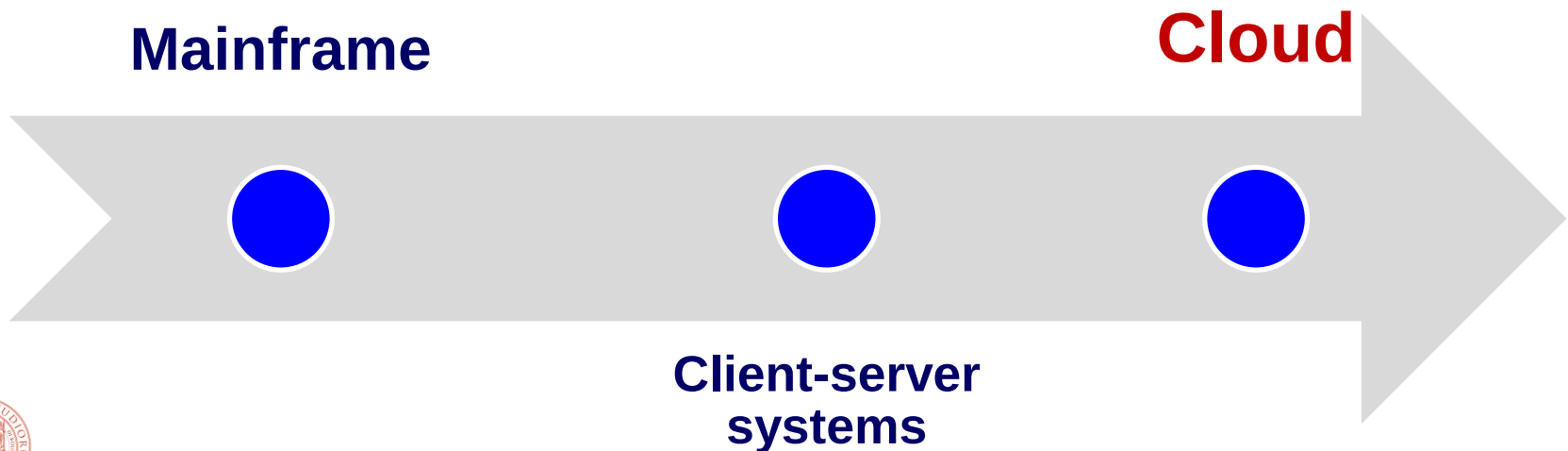
Cloud brings several disruptive changes

People	Process	Technology	Strategy	Structure
• Alignment of roles & responsibilities to service delivery • Staff training • Reorganization to adopt a service focus • Update of success metrics • Knowledge management	• Project planning • Capacity planning and compute resource procurement • Application prioritization • Developing & managing service levels • Technology Adoption (PoC, Pilot, Deploy) • End-user support	• Utility Computing architecture • Identity management • Data security • Data management • Systems management strategy • Vendor evaluation & implementation	• Defining a Cloud enabled IT strategy • Budgeting and project funding • Standards and guidelines • Reference architectures for SaaS, PaaS, and IaaS • Enterprise architecture	• Governance model • Defining and implementing controls • Identifying audit procedures



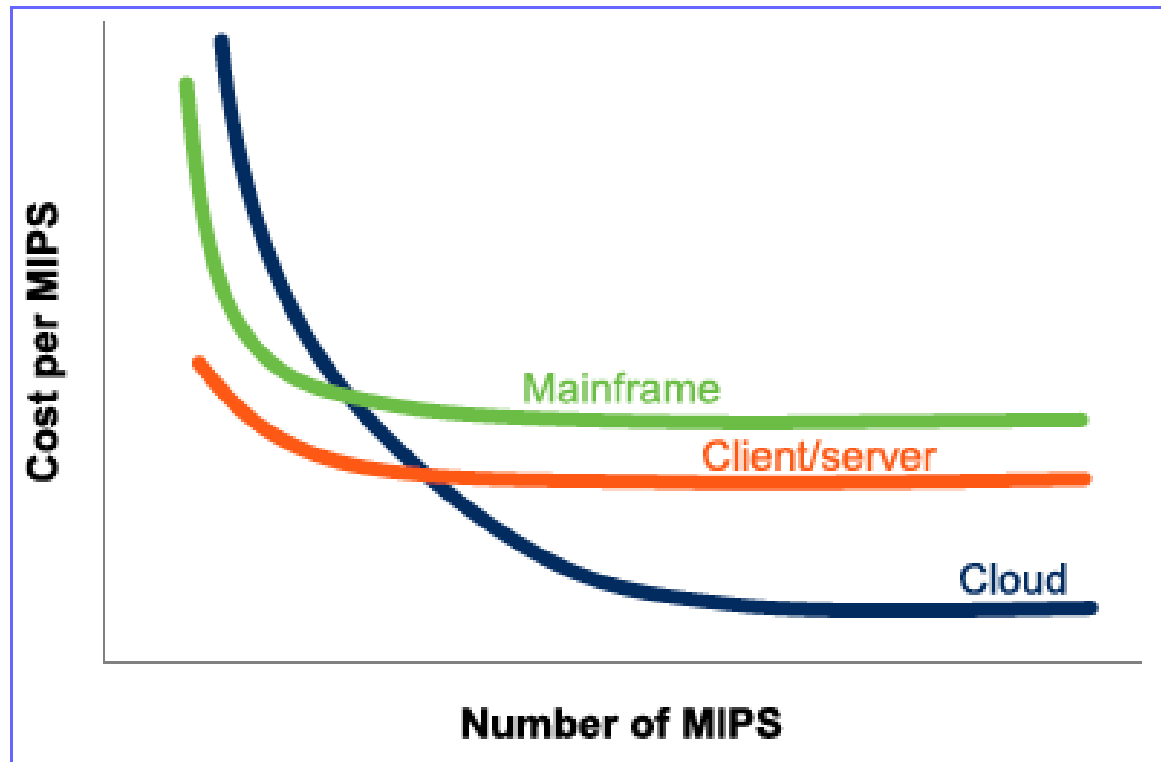
Evolution

- The mainframe era was characterized by significant economies of scale due to high up-front costs of mainframes and the need to hire competent personnel to manage the systems
- The client/server technology was characterized by high modularity and flexibility, and lower costs



Economies of scale

Cloud computing combines the best properties of mainframe (*economies of scale and efficiency*) and client/server computing (*modularity, agility, low costs*) thus eliminating one of the most famous trade-off of the IT history



IT revolution(s)



Two remarks

When a new technology appears, previous technologies survive (typically, for a long time).

Hybrid architectures will prevail in most companies

When a new technology appears,
you should NOT evaluate the pros and cons, and
you should NOT forecast the utilization models
looking just at *present* models and workloads
→ *New offers create new demand and new service requirements*



The emergence of cloud services is a fundamental shift of the economics of IT

- Cloud technology standardizes and pools computer/network resources and automates many of the maintenance tasks done manually in company data centers
- Cloud architectures facilitate elastic consumption, self-service, and pay-as-you-go pricing
- Cloud computing is not a return to the mainframe era, but it offers users economies of scale and efficiency that exceed those of a mainframe coupled with modularity and agility beyond what client/server technology offered



IT revolutions

- **First revolution**
 - To abstract the software from hardware
- **Second revolution**
 - To abstract business services from in-house built software



Winners and Losers

- **First revolution**



- **Second revolution**



If you want to win the challenge, build software that is ...

- **Agnostic** with respect to hardware, platform and software
- **Not agnostic** with respect to business processes

External customers and business people in your company
DO NOT CARE about your hardware/software choices or your preferred programming language, tool, operating system and libraries

They care about (low) COST and (short) TIME



Attributes

SERVICE VISION

Performance: response, latency, data loss

Reliability: hardware issues, replication

Scalability/Elasticity: more/less users, workload

Usability: compliance, operability

Security: user authentication and authorization

BUSINESS VISION

Customers

Values, Profit

Costs

Resources

Partners, Providers



Service attributes

Software *works well* independently of the number of users

Software *works always*

It is possible to increase/decrease software instances on demand and adopt a ***pay per use*** model

Performance

Scalability

Reliability

New business models



Darwin and adaptability

***“Its not the strongest of the species that survive,
nor the most intelligent,
but the ones most responsive to change”***

**For us, the best way to achieve ADAPTATION is through
SOFTWARE**

***Our “illusions” too are achieved through software
(virtualization is software)***



As in all good illusions, there are multiple “tricks”

We have exploited

- Increasing **computational** power (*thanks to Electronics*)
- Decreasing **communication** costs (*thanks to Telecommunications*)

We have created

- 1) Several layers of **virtualization**
- 2) Novel **programming** and **orchestration** paradigms
- 3) Novel **data management** paradigms
- 4) Novel **communication** paradigms



As in all good illusions, there are several “tricks”

We have exploited

- Increasing **computational** power and decreasing costs
(*thanks to Electronics*)
- Increasing **communication** power and decreasing costs
(*thanks to Telecom*)

Hardware

We have created

- 1) Several layers of **virtualization**
- 2) Novel **programming** and **orchestration** paradigms
- 3) Novel **data management** paradigms
- 4) Novel **communication** paradigms

Software



PROCESS

DATA

COMMUNICATION

The two faces of elasticity

- The workload may change rapidly
- The number of machines and resources you can get from a cloud provider can go *from one to virtually infinite and back*

Is your application able to scale-rescale similarly?

YES, if you use

- **Replicated processes that avoid:**
 - System and software bottlenecks
 - Single points of failure
 - Synchronous communications
- **(Autonomous) orchestration**



Old computing paradigms

Traditional, well-established paradigms do not work for cloud services based on large numbers of processes

Examples of “old computing paradigms” that does not satisfy modern service requirements

1. Large process synchronizations
2. Strong SQL-based data consistency
3. Store all information and then mine it *off-line*
4. Synchronous communications
5. All-to-all communications

PROCESS

DATA

COMMUNICATION



Cloud-based applications

- The biggest differences between cloud-based applications and traditional applications running in your data center are **reliability** and **scalability+elasticity**
- The cloud offers scalability and replication ***on demand***, thus allowing you to expand and contract your application as load fluctuates and/or failures occur
- **Elasticity and reliability attributes** are what makes the cloud really appealing, but they **cannot be achieved just by porting your existing application to the cloud**

In order to take advantage of what the cloud has to offer, you need to re-design your application around high scalability and reliability objectives



Re-design your application for scalability

1. Large parallelism

PROCESS

2. Asynchronous design of parallel processes

3. Limited inter-process communications

COMMUNICATION

4. Suitable (from *strong* to *lazy*) consistency
for data management

- Local distribution
- Geo distribution
- Internal caching
- External caching

DATA

**Everything related to enterprise's
business processes and organization**



CAVEAT for cloud PaaS users

1. **The consolidation of cloud services is a fundamental shift of the economics and pattern usage of IT** → Users already changed (“all Big Tech companies spoiled them”), hence all IT industry should change as well
2. **Scalability is well provided by the cloud.** What about your applications? → New paradigms are emerging
3. It is not guarantee that you save money when you use the cloud, but when you compare, include all the costs → **If you are not interested to availability and elasticity, then a cloud solution may be not convenient for you**
4. **When you entering the cloud, think in advance which is your *emergency exit (if it exists...)***



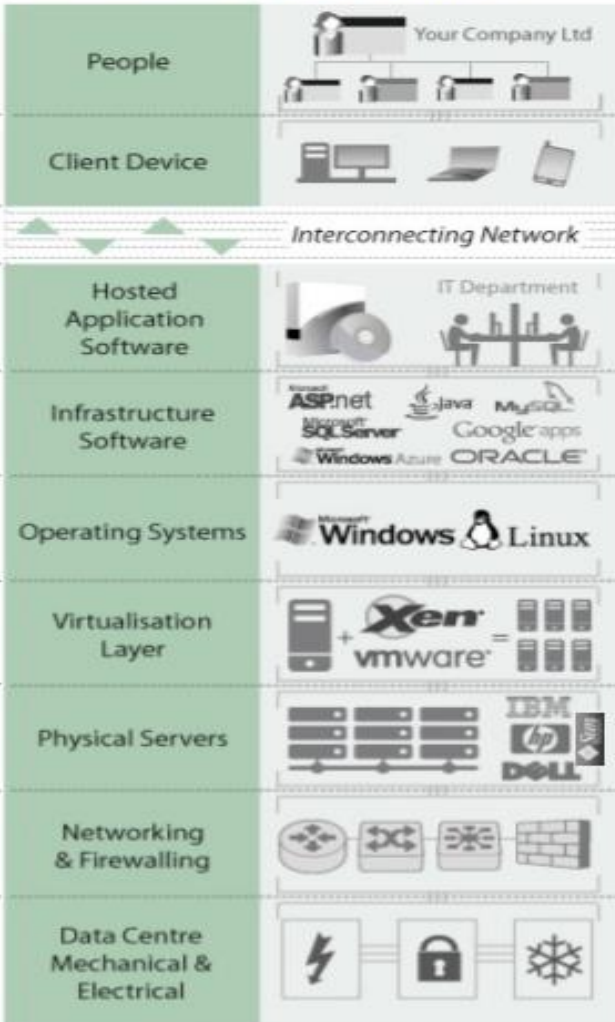
Not only IaaS-PaaS-SaaS



Today there are more than (3 service) x (4 deployment) paradigms

Service Layers Definition

Service stack components*



* as a Service

Moreover, the need to support cloud services integration and multiple clients require novel paradigms

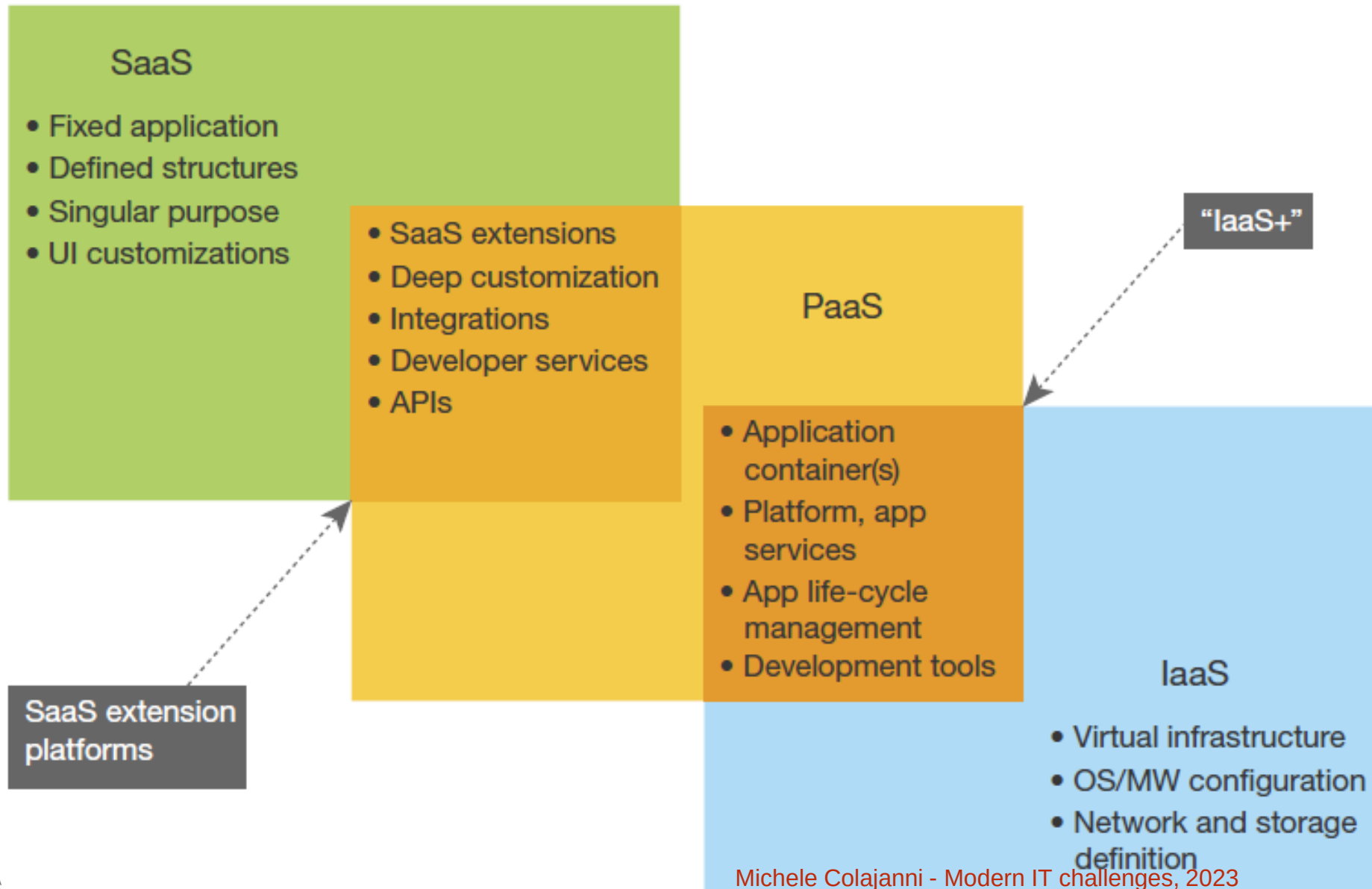
iPaas
aPaas
FaaS

Notes:

Brand names for illustrative / example purposes only, and examples are not exhaustive.

* Assumed to incorporate subordinate layers.

Disappearing boundaries



- Historical IaaS vendors (e.g., Amazon, Rackspace) are building services that deliver the abstraction and hosted middleware benefits that were often associated with PaaS
- Examples of IaaS+
 - AWS CloudFormation (create templates to describe AWS resources and dependencies)
 - Amazon Cognito (user management and permissions)
 - Elastic MapReduce (process vat amount of data)

Paas+IaaS

- Traditional PaaS vendors are reaching down to the virtual infrastructure
- Some PaaS originators now offer both highly abstracted platforms and transparency to the underlying plumbing
 - For example, Engine Yard provides both an abstracted interface for developers and the ability to configure underlying middleware, database, and IaaS resources
 - Heroku, one of Salesforce's cloud platforms, runs on AWS, and the vendor encourages developers to mix AWS and Heroku services



SaaS+PaaS

- SaaS vendors are pushing into platforms with extensibility features
 - **Salesforce's Force.com** platform is the leading example of a product that began life as a set of application extension tools and is now a public cloud platform used for many applications
 - Intuit, NetSuite, Workday, Box, and other SaaS vendors are beginning to follow, though most of them do not have complete PaaS offerings delivering independent and complete value for enterprises

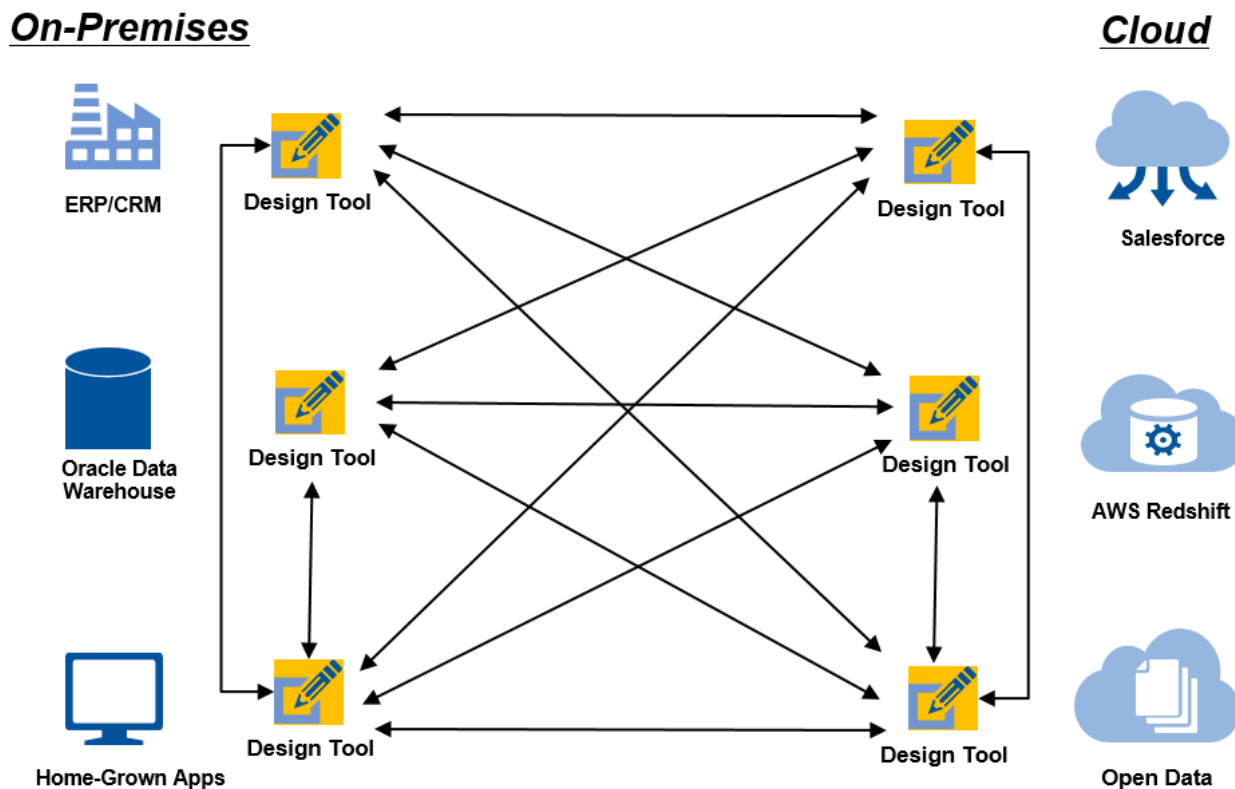


iPaaS



Hybrid architectures: cloud and on-premises

iPaaS is a suite of cloud services enabling development, execution and governance of integration flows connecting any combination of on premises and cloud-based processes, services, applications and data within individual or across multiple organizations



iPaaS functions

- **Basic iPaaS functions:**
 - platform for building and deploying integrations within the cloud and between the cloud and enterprise's services
 - iPaaS users can develop integration flows that connect applications residing in the cloud or on-premises, and then deploy them without installing or managing any hardware
- **Other iPaaS functions** include tools that support:
 - execution of integration data flows
 - development and life cycle management of integrated services
 - governance, management and monitoring of application and information flows
 - essential cloud features, such as cloud multi-tenancy and elastic provisioning



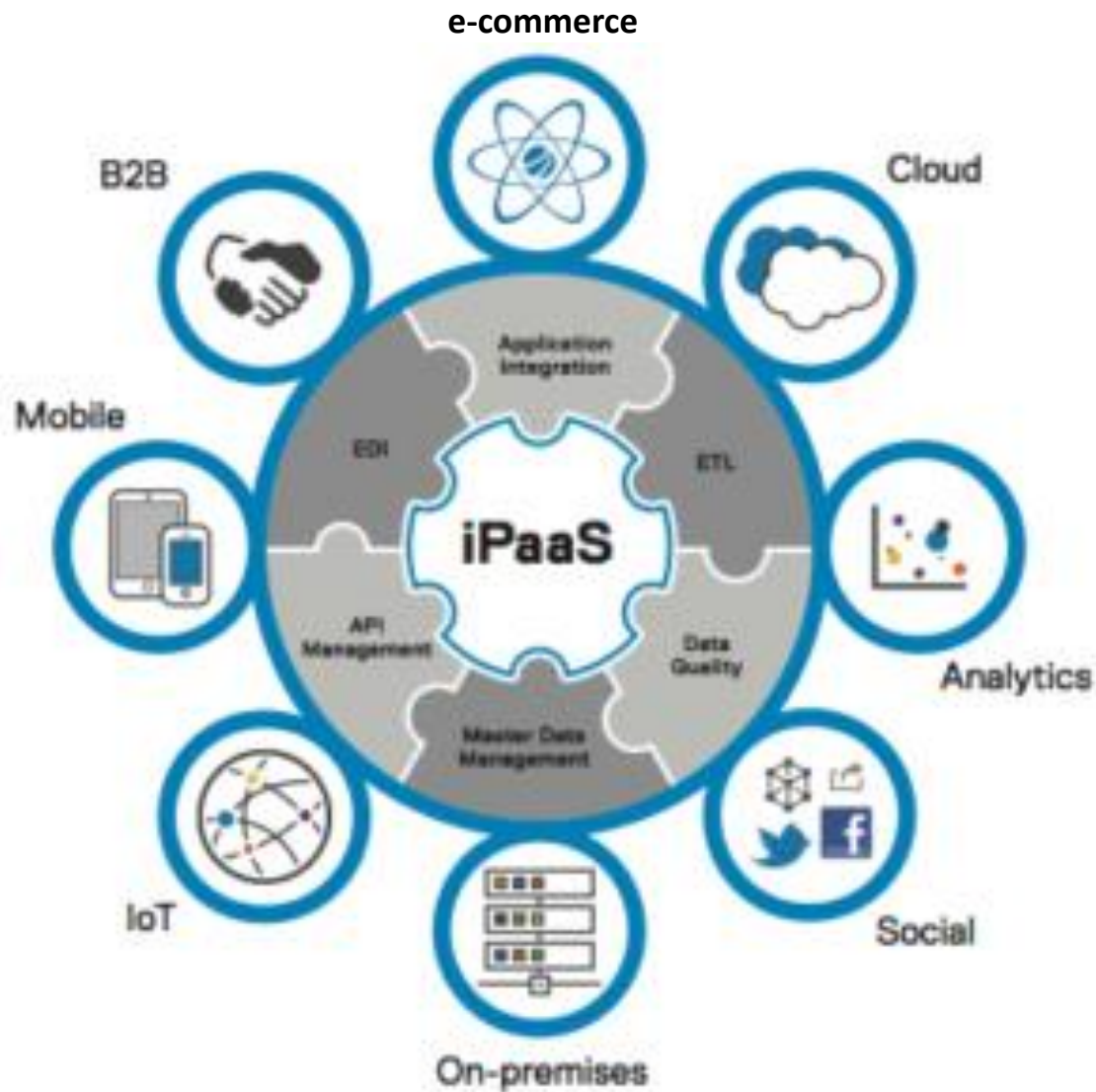
iPaaS vs SaaS

- **iPaaS** is a cloud-based integration solution, unlike SaaS which is hosted by one provider
- **iPaaS** allows applications that reside on-premise or on the cloud to integrate and have data to flow freely between them

This offering can bring different hosted software solutions



Possible applications



Example: Cloud Integration (1/2)

- Considering the mass migration of enterprises to cloud services, much of integration work involves the building of **cloud-to-cloud** and **on-premises-to-cloud** connections
- Traditionally, IT departments relied on **Enterprise Service Bus (ESB)** technology for application connections, but ESBs are not optimized for cloud-to-cloud and hybrid-cloud app integrations because they were born for managing complicated on-premises business systems and legacy messaging protocols
- You can use a modern **hybrid-ESB** for ground-to-cloud and cloud-to-cloud integrations, but setting up and managing hybrid-ESB connections is inefficient, time-consuming, and requires extensive skills and training



Example: Cloud Integration (2/2)

iPaaS functions can help in different ways:

- Large enterprises with complex, legacy infrastructures that want to incorporate new cloud strategies. iPaaS services for communications and big data integration can complement existing ESB infrastructure empowering on-premise connections and firewall-protected platforms with cloud-based solutions
- Businesses with cloud-based infrastructures. For simpler infrastructures that do not have to deal with antiquated messaging protocols or on-premise systems, an iPaaS can serve as the backbone to an entire cloud-based architecture
- iPaaS can offer solutions for upgrading an on-premise business intelligence systems to the world of real-time, big data, and machine learning analytics including the possibility of scaling and processing abilities to handle such tasks and integrating with cloud platforms for stream processing analytics



Example: API generation (1/3)

Data between business applications are integrated through APIs:

- They can connect the CRM software with enterprise accounting, inventory, data storage, human resources, and other applications
- They can also connect your applications with those of your partners, vendors, and clients

Hand-coding an API the traditional way is:

- a tedious and time-consuming process, taking about 34 business days for an in-house engineer to complete
- expensive; according to a famous developer the cost of building and testing one enterprise API is about \$15,000-\$25,000 (one order less in Italy...)



Example: API generation (2/3)

- 1) Some advanced iPaaS solutions offer a **no-code API generation interface** along with a library of pre-built connectors. This empowers users to build, configure, and publish APIs by pointing and clicking:
 - generate the API
 - create an RBAC (***Role-Based Access Control***) to granularly define **what a client can do with the API**
 - assign the RBAC to a generated API key
 - set restrictions on resource usage by assigning an access volume limit
- 2) If a native API connector is unavailable, you can get a scripting editor that supports PHP, Python, and NodeJS, thus letting advanced users develop new APIs that incorporate features like RBAC, API limits



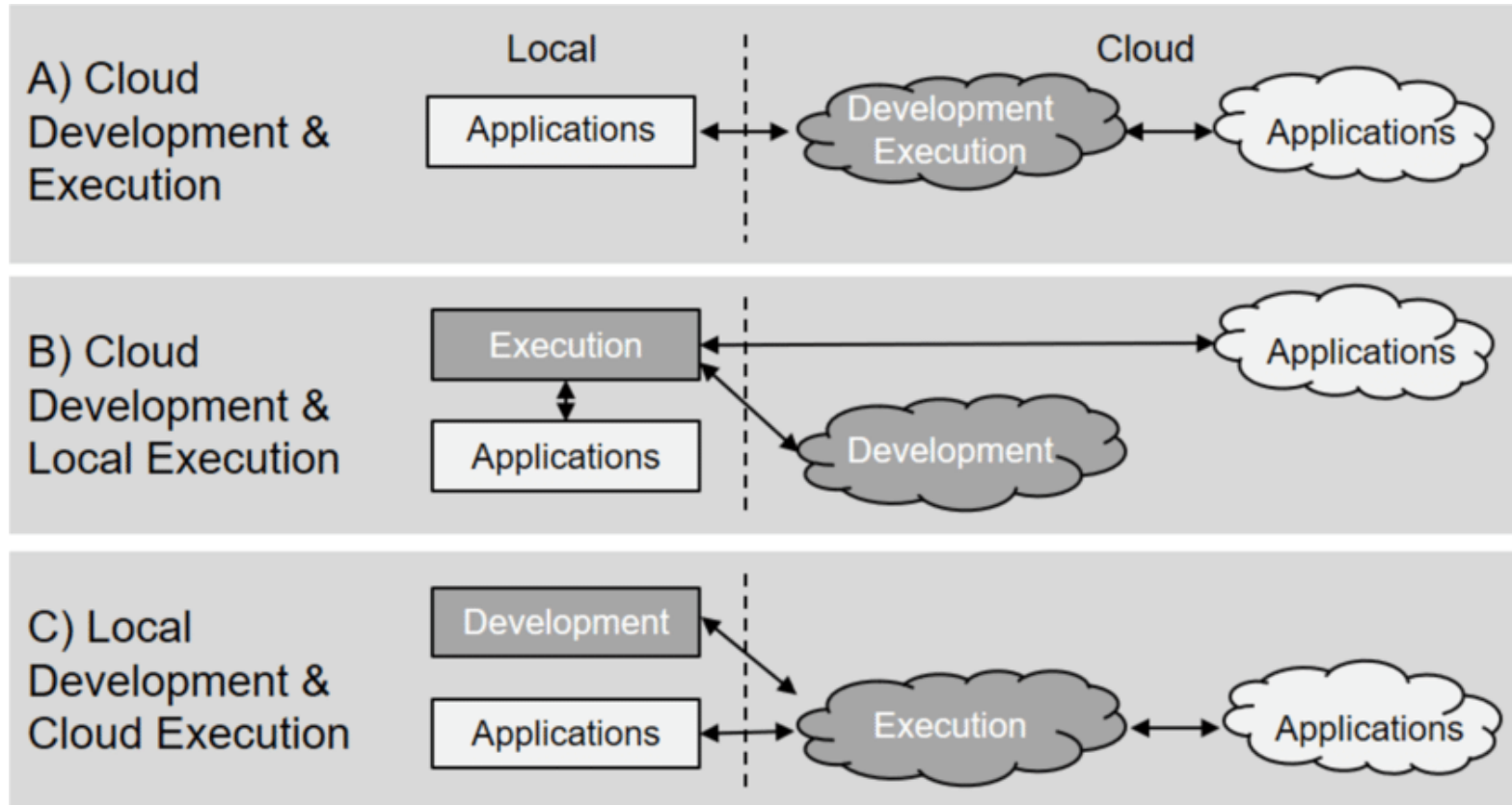
Example: API generation (3/3)

Benefits:

- Unburdens IT department
- Saves on outsourcing costs
- Speeds up your time to market
- Increases agility when onboarding new data, services, partners, applications, and technology



Possible iPaaS operations



Vendors: *leaders here are different*



Informatica (US): Integration data management

- Data Engineering
- Data Engineering Integration
- Data Engineering Quality
- Data Engineering Streaming
- Big Data Parser
- Edge Data Streaming
- Cloud Data Integration
- Cloud Application Integration
- API Management
- Cloud B2B Gateway
- Cloud Mass Ingestion
- Intelligent Structure Discovery
- Cloud Data Integration Elastic
- Data Security Cloud



aPaaS



Application platform as a service (aPaaS)

aPaaS is a cloud development environment characterized by:

- rapid application development
- low-code tools which substantially cut the amount of coding knowledge necessary to build apps

that:

- **brings the demanding art of programming applications to be accomplished even by non-developers**
- allows developers to deliver apps quickly so they can focus more on the creative aspects of app development
- augments productivity and reduces time to market
- improves operational agility that is, the ability to respond to changing operational conditions in terms of workload



The new class of “citizen developers”

- Low-code tools abstract away from code and offer tools to simplify application development → “**from Business Process Management to apps**”
- **Citizen developers** can create useful applications without vast expertise and coding knowledge
- In theory, assisted by low-code development tools, anyone can make a useful application
- aPaaS also offers opportunity to **expert developers** to create business applications faster, get them to market quicker, and without bogging down with repetitive tasks → projects that may take months or years to finish can be completed in weeks: savings can be as high as 350%, mostly due to reduced time to market for business applications



aPaaS

- **aPaaS** is a subcategory of PaaS, which exclusively offers cloud services to build apps, but do not offer other PaaS capabilities
- aPaaS solutions help small businesses transition to modern development, where agile, adaptive tools, and citizen developers are the traits for competitive evolution

Examples:

- WaveMaker
- Buddy
- Forms on fire
- Visual LANSA
- OutSystems
- Mendix
- ...



aPaaS vendors (1/2)

- **WaveMaker** is a cloud-based rapid application development platform which enables users to build Web and mobile apps and deploy these across multiple devices
- **Buddy** is a cloud-based DevOps automation platform for Web developers, which provides tools for building, testing, and deploying applications. The platform includes 100+ built-in actions to simplify development and deployment pipelines
- **Forms on fire** empowers users with an intuitive, easy to use and easy to support mobile cloud platform. Anyone can design using their Word & Excel tools for digitalizing forms and workflows
- **Visual LANSA** is a cloud-based application development solution designed to help software developers build to multi-experience applications for Web, desktop, and mobile devices. The WYSIWYG screen editor allows teams to improve any aspect of existing applications using one programming language



aPaaS vendors (2/2)

- **Restlet** is a provider of the Restlet Framework, an open-source API development framework that is a combination of APISpark (Restlet's specialized PaaS offering for building data-oriented Web APIs) and Restlet Studio (a free visual API design tool that accelerates the design of APIs through a lightweight Web-based development environment)
- **Mendix** offers
 - visual development, model driven (data models, UI, logic)
 - rapid Web-based prototyping through a WYSWYG editor
 - a collaborative development system for social collaboration, agile project management, team development, feedback management
 - reusability thanks to a public and private app stores, Micro services, page templates, building blocks, widgets



aPaaS vendors for real-time applications

- **PubNub** provides APIs for real-time application development
 - PubNub initiated as a provider of APIs that allowed publishers and developers to integrate messaging and social services on apps and websites
 - Now it offers PubNub Data Stream Network, a solutions that allows rapid development of real-time apps and communication for Web, mobile and Internet of Things (IoT)
- **Pusher** is a cloud service and provider of a hosted API for adding real-time bi-directional communication functionality to Web and mobile apps, or any Internet connected device
 - Example: a news website can use Pusher's API to add live delivery of curated content through social channels to create a real-time news feed



The present and future of application development

aPaaS is for non-expert or fast developers → small projects where it is important to have a faster way to build apps with high productivity

PaaS is for expert software designers and developers → any kind of project that can leverage of cloud platforms

iPaaS is for enterprise business applications → large projects that often have to integrate existing databases, data warehouses and applications (even on hybrid or multiple cloud platforms)



Enterprise data management with *open source* software

CLOUDERA



DATA CENTER &
PRIVATE CLOUD



HYBRID
CLOUD



MULTI
PUBLIC CLOUD

CLOUDERA
SDX

METADATA / SCHEMA / MIGRATION / SECURITY / GOVERNANCE



DATA
HUB



DATA FLOW &
STREAMING



DATA
ENGINEERING



DATA
WAREHOUSE



OPERATIONAL
DATABASE



MACHINE
LEARNING

CLOUDERA RUNTIME



CONTROL
PLANE



DATA
CATALOG



REPLICATION
MANAGER



WORKLOAD
MANAGER



MANAGEMENT
CONSOLE

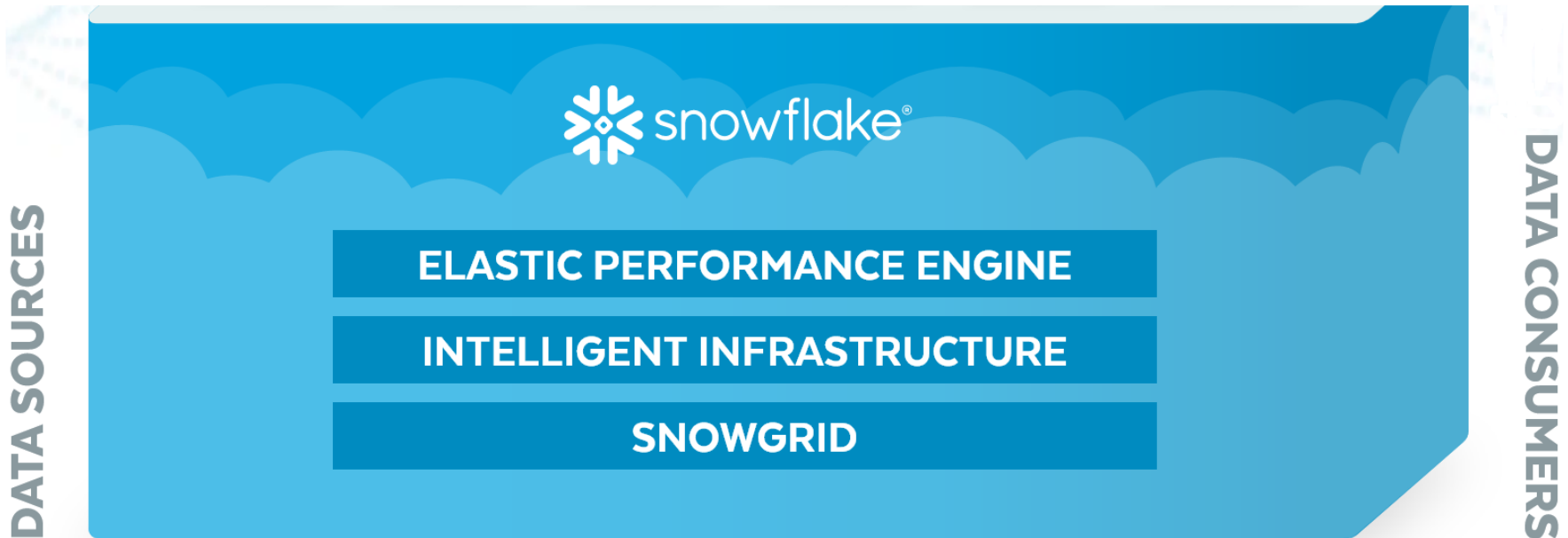


Cloudera stack (based on Apache tools)

- File system management: HDFS, Ozone
- Data collection and distribution processing: Hadoop
- Data engineering: Spark
- Data warehouse: HIVE (ITL), Impala (BI)
- Operational databases: noSQL HBase (enriched by Finix)
- Streaming analytics: Kafka and many others
- Machine Learning: Notebook and many others
- Project management: Git Hub, Maven, DevOps, ...
- Data visualization: Arcadia



Snowflake: oriented to Big data Management



1) Elastic Performance Engine

One engine that can power complex data pipelines, large-scale analytics, feature engineering, interactive applications across different and critical data workloads

- Instantly and cost-efficiently scale to handle any number of concurrent users and workloads, without impacting performance
- Software development
 - Robust SQL support
 - *Snowpark* developer framework (for Java and Scala accesses) that facilitates users even with low data management skills
 - Use of favorite external tools with native pushdown integrations



2) Intelligent Infrastructure

It is a fully managed service where automations help the customer to reduce risk, improve efficiencies, and focus on high-value data initiatives

- Snowflake can handle heavy lifting
- Snowflake is built for high availability and high reliability. It automates complex data replication, whether it is cross-region or even cross-cloud to recover quickly
- Snowflake gets continuous improvements for faster performance, better storage compression, and scalability — updates are all automatically with no action required
- **Costs:** usage-based pricing and fine-grained controls to rightsized resources for all types of data and workloads

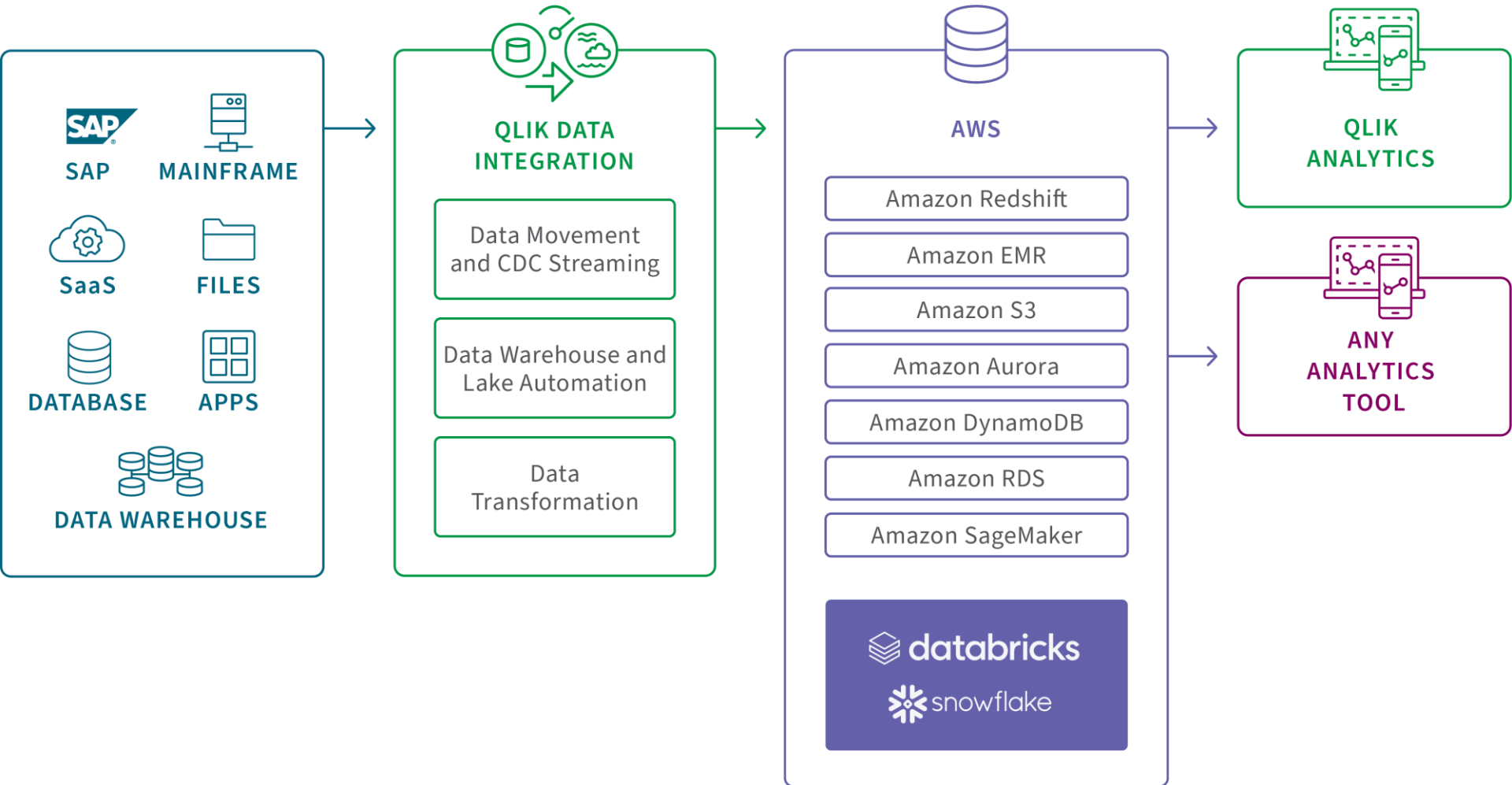


3) Snowgrid

- Share and collaborate easily without caring about the ETL process (Extract/Transform/Load), and with a single copy of data
- Break down data silos to instantly access data. Discover and share governed data between teams, business units, partners, customers
- Sophisticated cross-cloud governance controls and flexible policies ensure Snowflake's modern data sharing and collaboration is secure
- Policies follow the data to simplify governance at scale, while unlocking more value from even sensitive and regulated data
- Maintain new levels of business continuity with cross-cloud and cross-region connectivity, so to operate without disruption and deliver better customer experiences
- Compliance with regional and industry data regulations
- **Snowflake Data Marketplace**: discover and access the most relevant data, enrich insights with third-party data, connect with thousands of Snowflake customers, and extend workflows with data services and third-party functions



Integration



**Anything serious you do with
computers requires a *data
management strategy***

